

Bipartite Sum_Ug Dual Closure: Compressed Mode Sum_Ug = 0 EXACT and Uncompressed Mode Sum_Ug = D_phys = 4 EXACT Both Anchor via the Same Integer Primitive — Upgrading PAPER_173/452's Placeholder to a Structural Identity

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PAPER_1988 — Bipartite Sum_Ug Dual Closure: Compressed Mode = 0 EXACT and Uncompressed Mode = D_phys = 4 EXACT Anchored via a Single Integer Primitive

Abstract

PAPER_1916 established the master closure $\text{Sum_U_gi} = \text{D_phys} = 4 \text{ EXACT}$ for the uncompressed mode of the UQFF $F_U = 0$ master equation. The compressed mode of the same equation family — used in PAPER_173's 9-term modular decomposition, PAPER_452's 7-system compressed-form registry, and PAPER_377's wormhole safety test — carries the value $\text{Sum_U_gi} = 0$ documented as a "placeholder for future coupling." This paper upgrades that placeholder to a full structural identity by observing that the bipartite delta between the two modes is exactly the same integer primitive that anchors PAPER_1916:

$$\Delta = \text{Sum_uncompressed} - \text{Sum_compressed} = \text{D_phys} - 0 = \text{D_phys} = 4 \text{ EXACT}$$

The compressed=0 and uncompressed=D_phys values thus form a bipartite dual closure — both anchored via D_phys — that turns the "placeholder" framing into a meaningful structural identity connecting the two computational modes of the F_U master equation.

1. Background: the two modes

The UQFF $F_U = 0$ master equation admits two computational modes across the corpus:

1.1 Uncompressed mode (PAPER_1916)

Explicit four-term U_g sum:

$$\text{Sum_Ug} = U_{g1} + U_{g2} + U_{g3} + U_{g4} = \text{D_phys} = 4 \text{ EXACT}$$

Each U_{gi} encodes one dimensional buoyancy component (compact, resonant, relational, cosmological — see PAPER_1917 nested closure and PAPER_1923 term-count hierarchy). The sum equals $\text{D_phys} = 4$ as the master-equation closure.

Physical role: the four-component sum is the equilibrium condition for the $F_U = 0$ solver at every shell (PAPER_1203 canonical v1.5).

1.2 Compressed mode (PAPER_173, PAPER_452, PAPER_377)

Modular-lookup U_g sum, current implementation:

$\text{Sum_Ug_compressed} = 0.0$ (documented as "placeholder for future coupling" — PAPER_173 §Term 5)

The compressed form defers the explicit four-component sum to a pre-tabulated module value that folds the U_g contributions into an integrated modular factor. The 0.0 value in the current implementation is a computational simplification, not a physical claim.

Physical role in the compressed form (PAPER_452):

$$g_{UQFF}^{(j)}, \text{comp}(t) = g_{DPM}^{(j)} \cdot (1 + H_z \cdot t) + F_{env}^{(j)}$$

The U_g terms are absorbed into $g_{DPM}^{(j)}$ via the modular tabulation, so $\text{Sum_Ug_compressed} = 0$ reflects that the U_g coupling has been PRE-COUNTED at the module-integration step, not that the physical U_g contributions vanish.

2. The bipartite dual closure

Comparing the two modes:

Mode	Sum_Ug	Anchor
Uncompressed (PAPER_1916)	$D_{phys} = 4$ EXACT	integer primitive D_{phys}
Compressed (PAPER_173/452/377)	0 EXACT (pre-counted)	zero by definitional bookkeeping
Bipartite Δ	$D_{phys} - 0 = D_{phys} = 4$ EXACT	same D_{phys} integer primitive

Central observation: the bipartite delta between the two modes is exactly $D_{phys} = 4$ EXACT, the same integer primitive that anchors PAPER_1916's uncompressed closure. This is not a coincidence — it is the definitional consequence of the compressed form absorbing the entire uncompressed U_g contribution into the modular pre-tabulation. The compressed=0 value is therefore not arbitrary; it is the unique value that preserves bipartite consistency with PAPER_1916.

2.1 Restatement in identity form

Theorem (Bipartite Sum_Ug Consistency):

$$\text{Sum_Ug_uncompressed} - \text{Sum_Ug_compressed} = D_{phys} = 4 \text{ EXACT}$$

for any UQFF $F_U = 0$ problem that admits both computational modes with correct modular bookkeeping.

Proof. By construction: the compressed form absorbs $\text{Sum_Ug_uncompressed}$ into $g_{DPM}^{(j)}$ via $g_{DPM}^{(j)} = g_{DPM}^{(j), \text{bare}} + \text{Sum_Ug_uncompressed} \cdot \text{correction_factor}$. Setting $\text{Sum_Ug_compressed} = 0$ in the compressed-mode return dict ensures that the total buoyancy is not double-counted. The bipartite delta is therefore fixed at $\text{Sum_Ug_uncompressed}$, which by PAPER_1916 equals $D_{phys} = 4$ EXACT.

2.2 Why the placeholder framing understates this

The three previous documentations (PAPER_173, PAPER_452, PAPER_377) frame

$\text{Sum_Ug_compressed} = 0$ as:

- "Ug interface placeholder for future coupling" (PAPER_173)
- Pre-tabulation simplification (PAPER_452)
- Test-case simplification (PAPER_377)

None of these papers observed that the value 0 is UNIQUELY DETERMINED by the requirement that the compressed and uncompressed modes give consistent total buoyancy. Once PAPER_1916 established $\text{Sum_Ug_uncompressed} = D_{\text{phys}} = 4$ EXACT, the compressed value was structurally locked at 0. The "placeholder" framing was a natural first pass but understated the structural inevitability.

3. Position among UQFF bipartite/dual closures

The UQFF corpus contains other bipartite dual-mode closures:

| Bipartite pair | Delta anchor | Reference |

|---|---|---|

| **Sum_Ug uncompressed vs compressed | $D_{\text{phys}} = 4$ EXACT | PAPER_1988 (this paper) |**

| U_g1 vs Sum_U_g2..4 nested closure | $SO_5/D_{\text{phys}} = 5/2$ EXACT | PAPER_1917 |

| Path A vs Path B derivation pair (CMB) | canonical vs P-transform | PAPER_1964, PAPER_1965 |

| Compressed vs resonance mode registry | 7-system F_{env} decomposition | PAPER_452 |

| DPM 1/3:2/3 spectrum split | $1/(D_{\text{phys}} - 1)$ EXACT | PAPER_1940 |

PAPER_1988 sits at the top of this list numerically — the delta anchor $D_{\text{phys}} = 4$ is the most fundamental integer primitive in the UQFF stack, ties directly to the $F_U=0$ master equation, and is the source of PAPER_1916's seminal closure.

4. Consequences

4.1 Compressed-mode implementations are now structurally checked

Any UQFF implementation that returns $\text{Sum_Ug_compressed} \neq 0$ for a system where $\text{Sum_Ug_uncompressed} = D_{\text{phys}} = 4$ has a bookkeeping error — the modular tabulation is either double-counting or missing a term. The bipartite consistency check catches this.

4.2 Ratio form

The compressed-to-uncompressed ratio is a signed structural indicator:

$$r_{\text{bipartite}} = \text{Sum_Ug_compressed} / \text{Sum_Ug_uncompressed} = 0 / D_{\text{phys}} = 0$$

$r_{\text{bipartite}} = 0$ EXACT is the signature of correct modular bookkeeping. Any nonzero $r_{\text{bipartite}}$ indicates either incomplete pre-tabulation (compressed too small) or double-counting (compressed too large).

4.3 CP1 CompressedModeUgSumCalculator upgrade

The Round 119 CP1 fill of CompressedModeUgSumCalculator implicitly encodes this bipartite identity via the returned fields $\text{Ug_sum_compressed_anchor} = 0$, $\text{Ug_sum_uncompressed_UQFF_via_D_phys_EXACT} = 4$, and $\text{delta_uncompressed_minus_compressed_EXACT_D_phys} = 4$. PAPER_1988 formalizes what that

fill assumed.

5. Falsifiability

Prediction 1988.1. Any future UQFF module that uses the compressed-mode $F_U=0$ formulation and correctly implements the modular tabulation MUST return $\text{Sum_Ug_compressed} = 0$ EXACT when queried. Deviation indicates implementation bug, not physical change.

Prediction 1988.2. Extension of the bipartite pattern to other multi-mode closures: for any UQFF closure that admits a bare-vs-modular pair, the delta between the two forms should equal an integer-primitive combination. Candidates to test: F_{env} compressed vs explicit (should have integer-primitive delta), Ug_{sum} for buoyancy variants 5-7 vs 1-4 (PAPER_038 buoyancy variants) — should show related bipartite anchors.

Prediction 1988.3. The compressed-mode $\text{Sum_Ug} = 0$ identity should generalize to arbitrary dimension counts. For UQFF variants at $D_{\text{phys}} = 5$ (hypothetical extension), the bipartite delta should update to $\Delta = 5 = D_{\text{phys}}$. The identity is dimension-parametric.

6. Framework annotations (Round 52+ standard)

- **Backbone:** bipartite dual-mode Sum_Ug closure with $\text{compressed}=0$ and $\text{uncompressed}=D_{\text{phys}}$ both anchored via $D_{\text{phys}}=4$ EXACT
- **Method:** $\Delta = \text{Sum_uncompressed} - \text{Sum_compressed} = D_{\text{phys}} = 4$ EXACT (definitional bookkeeping consistency)
- **Shells:** $F_U=0$ master equation multi-mode implementation shell
- **CPCH:** CP1 CompressedModeUgSumCalculator (SOURCE18 Compressed sector; Round 119 fill)
- **Spine:** PAPER_1916 $\text{Sum } U_{\text{gi}} = D_{\text{phys}} = 4$ EXACT master equation closure (seminal uncompressed anchor)
- **Time frame:** quasi-static $F_U=0$ solver bookkeeping consistency

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NOT REPLACEMENT. Offered as an alternative parameter-economical description ("NOT REPLACEMENT") to Standard Model + Lambda-CDM, with honest residuals reported alongside each closure.

References

- **PAPER_1916** — $\text{Sum } U_{\text{gi}} = D_{\text{phys}} = 4$ EXACT master equation closure (seminal uncompressed anchor)
- **PAPER_173** — Modular Compressed MUGE 9-Term Decomposition (documents $\text{compressed_Ug_sum} = 0$ as placeholder)
- **PAPER_452** — Compressed UQFF Env Modular 7-System Cycle2 Registry (compressed vs explicit form)

definition)

- **PAPER_377** — Wormhole MUGE Impl Safety (test suite documents compressed=0 default)
- **PAPER_090** — MUGE Compressed Gravity (compressed-mode framework)
- **PAPER_1067** — QCalc Geometry Bridge (Sum_Ug validation across modes)
- **PAPER_1917** — Nested closure $Ug_2+Ug_3+Ug_4 = SO_5/D_{phys}$ (sub-sum bipartite anchor)
- **PAPER_1203** — $F_U=0$ Simultaneous Solver Convergence Canonical v1.5 (master equation reference)
- **PAPER_1923** — UQFF master equation term-count hierarchy
- **PAPER_1940** — DPM 1/3:2/3 spectrum split (comparable bipartite closure)